One Operator, Many Physics, Several Shapes

A Synthesis of the KSF (Kosaka Skin-o-hedron) Notebook

Abstract

A single discrete operator $L = M^{-1}K$, built once from the cotangent / P1 finite-element stiffness K and lumped mass M on an equally-spaced tetrahedral mesh (the *Kuhn cube*), is shown—to machine precision against exact solutions—to reproduce heat diffusion, scalar-wave propagation, damped and boundary-coupled dynamics, and the pressure field of a fluid, and to carry those same physics onto reshaped domains (torus, sphere). The discipline throughout is *kūron o kasanenai* (*do not stack unverified claims*): nothing advances until a runnable check matches a known answer. The mathematics is classical; the contribution is a single, self-consistent, end-to-end *verified* realisation of all of it on one operator. Every figure in Appendix A is the live output of a script in `src/verification3d/`.

1 Thesis

A single discrete operator,

$$L = M^{-1}K,$$

drives every physics in this notebook. It is not new mathematics: the cotangent Laplacian, finite elements, periodic boundary conditions, and the curvature obstruction are classical. What is here is a *single, self-consistent, end-to-end verified* realisation of all of them on one operator, each link confirmed by its own exact-solution test.

2 The one idea

On each tetrahedron a P1 (linear) basis gives a constant gradient operator G (3×4). Assembling $K = \sum_{\text{tet}} \text{vol} \cdot G^{\mathsf{T}}G$ yields the stiffness; lumping volumes gives the diagonal mass M. Then

$$\text{heat:} \quad \frac{\partial T}{\partial t} = -L\,T, \qquad \text{wave:} \quad M\ddot{p} = -Kp, \qquad \text{pressure:} \quad Kp = Df,$$

with decay $e^{-\lambda t}$ and oscillation $\cos(\sqrt{\lambda}\,t)$ on the *same* spectrum, and a single Poisson solve for pressure whose gradient and divergence are literally the operator inside K ($D \circ \text{grad} = K$). The spectrum of L is the shared backbone: heat rides it as decay, waves as oscillation, pressure as its static and acoustic limit.

3 The arc

Step 1 — one operator from two routes. A wave-propagation (S-parameter) route and a heat-conduction route are shown to be the same L. Its lowest Neumann eigenvalues match $\pi^2 = 9.8696$ (measured $9.72, 9.86, 9.86$); a two-material interface reproduces the exact contact value $T = k_2/(k_1 + k_2)$ to 2×10^{-15} and below; heat energy decays as $e^{-2\lambda t}$ (measured 0.6168 vs predicted 0.6170); the wave solver is stable (energy fluctuation 1.2×10^{-2}). The mesh's only blemish, a residual ∼5% directional anisotropy $\Delta c/c \approx 0.047$–0.048, is the Kuhn cube's broken symmetry, inherited honestly everywhere downstream.

Step 1.5 — dynamics and boundaries. Damping $M\ddot{p} = -Kp - \gamma M\dot{p}$ produces the predicted envelope (γ=0.5 tail/head 0.101 vs 0.092; γ=2 $\to 10^{-4}$). A discrete dispersion cutoff appears at $f \approx 0.127$ cycles/step. Robin convection $q = h\,B\,(T - T_{\text{env}})$ gives half-life $\propto 1/h$ (the product half-life $\times\, h$ holds at 3077–3144 across $h = 0.5 \ldots 8$, a 2.2% spread). Stefan–Boltzmann radiation $q = \varepsilon\sigma B(T^4 - T_{\text{env}}^4)$ shows the steep T^4 signature.

Step 2 — pressure as one Poisson solve. The gradient/divergence assembled from the same per-tet G satisfy $D(\operatorname{grad} p) = Kp$ to 3.4×10^{-16}. Five exact targets pass: hydrostatic $\operatorname{grad} p = \rho g$ to 3.6×10^{-11}; Helmholtz projection (pure gradient removed to 3.8×10^{-15}, idempotent to 2.9×10^{-14}); the gas/liquid split (exponential atmosphere vs linear law, joined by compressibility c^2); the gas acoustic limit reproducing Step 1's wave exactly ($\omega = c\sqrt{\lambda}$, relative error 0.000); and the pure-Neumann null space ($|K \cdot \mathbf{1}| = 6 \times 10^{-15}$, compatibility $|\mathbf{1} \cdot b| = 7 \times 10^{-15}$). **Step 2 contains Step 1** as its acoustic limit.

V2.5 — reshape the domain. A mesh has connectivity and geometry, behaving oppositely under deformation. *Topological* deformation (gluing opposite faces, no vertex moves) preserves L exactly: the 3-torus folds 729 $\to$ 512 vertices, its spectrum matches the analytic flat-torus $(2\pi)^2 = 39.478$ (measured 37.49, a 5% FE error) with the correct **6-fold degeneracy** and one constant null mode, and the seam is seamless (every vertex has identical degree 15). *Geometric* deformation (moving vertices) degrades L: a cube$\to$ball map drops tet quality from 0.657 to 0.014 and creates 2844 negative-conductance (sign-flipped) cotangent edges; a wrapped torus, 2726. The physics still solves on the degraded mesh. And there is a first-principles reason the torus is exact but the sphere is not: the *torus is flat* (a periodic box) while the *sphere is curved*, so by the *Theorema Egregium* no distortion-free cube$\to$sphere map exists (best sampled quality 0.224, never the baseline 0.657).

The sign-flip question (§5b) — settled honestly. Can a different weight formula (norm) remove those negative conductances? Not for free. The sign-flips come exactly from **obtuse dihedral angles** (the Kuhn cube sits at 90° precisely, so any deformation tips some angles obtuse: 816 flips $\leftrightarrow$ 1260 obtuse on the n=6 ball). Swapping the cotangent weight for a uniform (graph) or clamped weight removes the flips but destroys the linear-exactness the verification rests on (harmonic reproduction of $u = x$ jumps from 2.8×10^{-16} to 2–9 $\times 10^{-2}$). Quality-guarded smoothing raises quality (0.401 $\to$ 0.513) and keeps exactness, yet does *not* remove the flips (816 $\to$ 838): a flip is an edge-summed property and globally non-obtuse 3D meshes are genuinely hard. Only geometry preservation gives both at once: the undeformed cube has 0 flips *and* machine-precision exactness. The clean path is topological deformation, or accepting graceful degradation—never a cheaper norm.

Grand finale — made visible. A single viewer (`viewer/viewer_solid.html`) runs all of it: three solids (cube, sphere, torus) $\times$ four physics (heat, scalar wave, liquid, gas), every displayed value from the verified Python solver (preset or live server solve), with wireframe, transparency, and slice.

4 Why this matters (and its honest scope)

That the framework *transforms*—the same operator simulating the same physics on different shapes—is the suggestive part: it is the property a simulator needs to apply to varied engineered geometries, not just a single test box. Stated plainly:

- **Not new theory.** Textbook discrete differential geometry and FE analysis; no novel mathematics is asserted.

- **Genuinely verified, end to end.** One operator carried from heat to waves to pressure to shape-change, each link confirmed against an exact answer at 10^{-11}–10^{-16}, including the negative results (the sign-flip study, the sphere obstruction).
- **Bounded reach.** A linear, scalar-centred, n=8 structured-grid notebook—not a commercial CAE solver. Correct, verified, and transparent within its reach; viscous advection, true curved-manifold operators, and optimal remeshing are out of scope.

The trajectory named by the author—*from "no free lunch" to "a proper lunch at a good restaurant"*—is apt: there is no free lunch, but paid for correctly (topological transform, or measured degradation), it yields a real, verified meal.

A Verification results

Each row is asserted by the named script; *measured* values are live output.

Step 1 — `unified_scalar_verify.py`

claim	exact target	measured	status
Shared spectrum (Neumann)	$\pi^2 = 9.8696$	$9.72, 9.86, 9.86$	PASS
Interface k_1:k_2 = 1:1	$T = 0.5$	err 2.0e−15	PASS
Interface 1:3 / 1:10	0.75 / 0.909	9.1e−15 / 3.3e−16	PASS
Heat energy decay	$e^{-2\lambda t} = 0.6170$	0.6168	PASS
Wave stability	bounded	fluct. 1.2e−2	PASS
Mesh anisotropy (blemish)	—	$\Delta c/c \approx 0.047$–0.048	noted

Step 1.5 — `dynamics_boundary_verify.py`

claim	exact target	measured	status
Damped envelope γ=0.5	0.092	0.101	PASS
Damped envelope γ=2.0	$\sim 10^{-4}$	10^{-4}	PASS
Dispersion cutoff	band ~ 0.13	0.127 cyc/step	PASS
Robin half-life $\propto 1/h$	const half $\cdot\, h$	3077–3144 (2.2%)	PASS
Stefan–Boltzmann T^4	steep cooling	many-fold faster	PASS

Step 2 — `pressure_field_verify.py`

claim	exact target	measured	status
Grad/div consistency	$D(\mathrm{grad}\, p) = Kp$	3.4e−16	PASS
Hydrostatic grad $p = \rho g$	exact	3.6e−11	PASS
Projection: gradient removed	0	3.8e−15	PASS
Projection idempotent P^2=P	0	2.9e−14	PASS
Gas vs liquid (c^2)	exp $\leftrightarrow$ linear	gap 8.81 $\rightarrow$ 4.8e−11	PASS
Acoustic limit $\omega = c\sqrt{\lambda}$	exact	rel 0.000	PASS
Null space / solvability	0	6e−15, 7e−15	PASS

V2.5 — `mesh_transform_verify.py`

claim	exact target	measured	status
Topological 3-torus fold	$729 \to 512$	512	PASS
Flat-torus spectrum	$(2\pi)^2 = 39.478$	37.49 (5%)	PASS
Degeneracy / null	6-fold, 1 null	6, 1	PASS
Seamless wrap	uniform degree	15 everywhere	PASS
Geometric ball distortion	—	$q_{\min}$ 0.014, 2844 flips	PASS
Geometric torus distortion	—	q_{mean} 0.053, 2726 flips	PASS
Topological torus distortion	preserved	q 0.657, 0 flips	PASS
Sphere obstruction (Egregium)	$<$ baseline	best $q_{\min}$ 0.224	PASS
Physics on degraded mesh	bounded	wave ratio 0.337	PASS

§5b — `cotangent_signflip_verify.py`

claim	exact target	measured	status
Mechanism: flips $\leftrightarrow$ obtuse	cube 0/0	ball 816/1260	PASS
cotangent exact but flips	$\sim$machine	2.8e−16, 816	PASS
uniform/clamp flip-free, inexact	flips 0	err 2–9e−2	PASS
Smoothing: quality up, exact kept	$q\uparrow$, exact	0.401→0.513	PASS
Smoothing keeps flips	honest	$816 \to 838$	PASS
Clean escape (geometry kept)	0 + exact	0, 3.33e−16	PASS

B Map of the notebook

Theory (`theory/`): `01`–`01e` S-parameter wave route; `02` heat route; `03` unified scalar operator; `03b` dynamics & boundaries; `04` pressure; `05`(+`5b`) mesh transform; this `00` synthesis. **Verification** (`src/verification3d/`): `unified_scalar_verify`, `dynamics_boundary_verify`, `pressure_field_verify`, `mesh_transform_verify`, `cotangent_signflip_verify`, plus S-parameter / heat checks and the viewer exporters. **Core** (`src/ksf3d/`): `mesh3d_uniform.py` (`kuhn_cube`), `fem3d.py` (`fem_laplacian`). **Viewers** (`viewer/`): `viewer_unified.html`, `viewer_pressure.html`, `viewer_solid.html`, each with presets and an optional server-solve API.

Every figure in Appendix A is the live output of the corresponding script at the time of writing; re-running them reproduces it.